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Ontology-Driven Validation Framework for Building Pathology Diagnosis: A Deterministic Element–Symptom–Pathology Model for Data Integrity in Public Asset Management

A Semantic Data Integration Method for Structured and Reliable Inspection Data in Digital Twin Environments

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ABSTRACT

The management of public building assets is frequently hindered by fragmented inspection practices and inconsistent diagnostic records, largely driven by subjective interpretation and the absence of structured data validation mechanisms. This limitation compromises the reliability of asset condition data and restricts its effective use in evidence-based decision processes.

This paper presents an ontology-driven validation framework for building pathology diagnosis, developed within the PPMI Digital Twin system for the public infrastructure of the Autonomous Region of the Azores. The proposed model introduces a deterministic semantic structure based on a hierarchical validation chain — Element–Symptom–Pathology — which constrains user input through predefined dependency relationships. This approach ensures that each diagnostic record is logically consistent, eliminating semantically invalid or technically incompatible data at the point of acquisition.

Unlike conventional inspection systems that rely on unrestricted input and user interpretation, the proposed framework embeds engineering knowledge directly into the data collection process through a controlled ontology and a closed-loop validation mechanism. The ontology defines a structured vocabulary of building elements, observable symptoms, and correlated pathologies, establishing explicit semantic relationships that reflect real-world technical dependencies.

The implementation of this model within an operational environment demonstrates a significant improvement in data consistency, traceability, and reliability. By enforcing deterministic validation, the system transforms inspection data into a structured and interoperable information asset, supporting scalable data governance in public asset management contexts.

The results highlight that ontology-based validation frameworks can serve as a foundational layer for reliable public asset data, enabling the transition from subjective inspection practices to formalize, knowledge-driven processes. Furthermore, the structured datasets generated by this approach provide a robust basis for future integration with advanced analytical and machine learning systems.

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1. INTRODUCTION

The effective management of public building assets depends on the availability of reliable, structured, and semantically consistent data regarding their physical condition. However, current inspection practices are frequently based on subjective assessments and unstructured data input, resulting in inconsistent diagnostic records and limited interoperability across digital systems. This lack of structured validation leads to semantic ambiguity, diagnostic inconsistency, and reduced confidence in the resulting datasets, which is a widely recognized limitation in facility management practices (Zhao et al., 2022).

Consequently, the collected information becomes difficult to standardize, aggregate, and integrate into broader asset management frameworks such as Building Information Modelling (BIM) environments. The absence of formalized relationships between building elements, observable symptoms, and associated pathologies represents a critical limitation in the transition towards data-driven public asset management. Without controlled vocabulary and a validation mechanism, the integrity of inspection data cannot be guaranteed, compromising decision-support processes and long-term asset management strategies. To address these challenges, this paper proposes an ontology-driven validation framework for building pathology diagnosis.

The proposed model introduces a deterministic validation approach based on a structured relationship between three core entities: Element, Symptom, and Pathology. By enforcing a hierarchical and dependency-based validation chain, the framework constrains data input and ensures that only semantically consistent and technically valid combinations are recorded. This approach is implemented within the context of the PPMI Digital Twin, a structured information system developed for the Regional Government of the Azores to support the management and monitoring of public building assets in the region. The system provides an operational environment for standardized inspection procedures, enabling the application of the proposed ontology-driven validation framework and supporting consistent and reliable data acquisition.

The ontology serves as the semantic backbone of the system, enabling standardized inspection procedures, improving data reliability, and supporting scalable public asset data governance. The main contribution of this work lies in demonstrating how a deterministic, ontology-based validation model can transform building pathology diagnosis from a subjective and descriptive process into a structured and knowledge-driven system. By ensuring data integrity at the point of acquisition, the proposed framework establishes a foundation for reliable public asset data and future integration with advanced analytical and intelligent systems.

2. PROBLEM STATEMENT

The diagnosis of building pathologies in public asset management systems is frequently affected by the lack of structured data validation mechanisms and the absence of formal semantic relationships between inspection variables. In current practice, inspection processes typically allow unrestricted associations between observed symptoms and identified pathologies, relying heavily on individual interpretation and professional experience. This approach introduces significant inconsistencies in diagnostic records, as similar physical conditions may be described differently across inspections, while distinct technical issues may be incorrectly classified under the same pathology. The absence of controlled vocabularies and dependency-based validation results in datasets that are semantically ambiguous and difficult to standardize, a limitation also highlighted in studies addressing data integration challenges across the building lifecycle (Deng et al., 2021).

The lack of a deterministic validation framework prevents the enforcement of technically coherent relationships between building elements, observable symptoms, and their underlying causes. Without a structured mechanism to constrain possible combinations, inspection data may

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include incompatible or invalid associations, compromising their reliability and limiting their usability in systematic analysis and asset management processes. These limitations create a fundamental barrier to the establishment of reliable public asset data, as fragmented and inconsistent information structures hinder the effective transition from BIM-based models to integrated digital representations.

Inconsistencies and semantic ambiguities undermine the comparability of records, reduce confidence in diagnostic outputs, and restrict the integration of inspection data into structured information systems. Therefore, there is a clear need for a validation framework that enforces logical consistency, ensures semantic coherence, and supports the generation of reliable datasets. Such a framework must be capable of structuring the relationships between inspection variables and constraining data input through a deterministic approach, thereby eliminating ambiguity and enhancing the overall quality of diagnostic information.

3. Methodology

This section presents the ontology-driven validation framework proposed for building pathology diagnosis. The framework is based on a deterministic semantic model that structures the relationships between building elements, observable symptoms, and associated pathologies, ensuring data integrity through controlled input and dependency-based validation.

3.1 Conceptual Model

The core of the proposed framework is defined by a hierarchical relationship between three fundamental entities: Element, Symptom, and Pathology. These entities represent, respectively, the physical component of the building, the observable manifestation of a condition, and the underlying technical cause. The model is structured as a directed and ordered validation chain:

Element–Symptom–Pathology (deterministic validation chain)

In this structure, the selection of each entity directly constrains the possible values of the subsequent entity. This dependency-based sequence establishes a deterministic validation process, eliminating ambiguity and enforcing logical consistency across all diagnostic records, in line with integrated digital twin data models that rely on structured relationships between physical and virtual assets (Tao et al., 2018).

3.2 Ontology Structure

Ontology is structured around a controlled set of entities and predefined relationships that reflect technical dependencies observed in building pathology diagnosis. The framework is based on three core components: Element, Symptom, and Pathology. Each Element is associated with a specific set of observable Symptoms, and each Symptom is linked to a limited set of technically consistent Pathologies. These relationships are not arbitrary but are defined based on engineering knowledge and validated inspection practices. The ontology operates as a controlled vocabulary in which all possible combinations are predefined and constrained within the system. This ensures that users are guided through a structured inspection process, where each selection is limited to technically valid options, a principle aligned with digital twin architectures that integrate heterogeneous models through structured data relationships (D'Amico et al., 2019).

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3.3 Deterministic Validation Mechanism

The validation process is enforced through a structured and sequential workflow in which user input is constrained at each stage of the inspection procedure. The process begins with the definition of the spatial context of the inspection, followed by the application of the ontology-driven validation chain. The workflow operates in two main stages:

a) Spatial Context Definition

The user first identifies the location of the inspection within the building asset by selecting:

Building–Block–Floor–Macrozone–Room

This hierarchical structure ensures that each inspection record is spatially contextualized within the asset. It defines the spatial context of the inspection, serving as a prerequisite for the ontology-driven validation process. The proposed framework is illustrated in Figure 1, which integrates the spatial context definition with the ontology-driven validation chain.

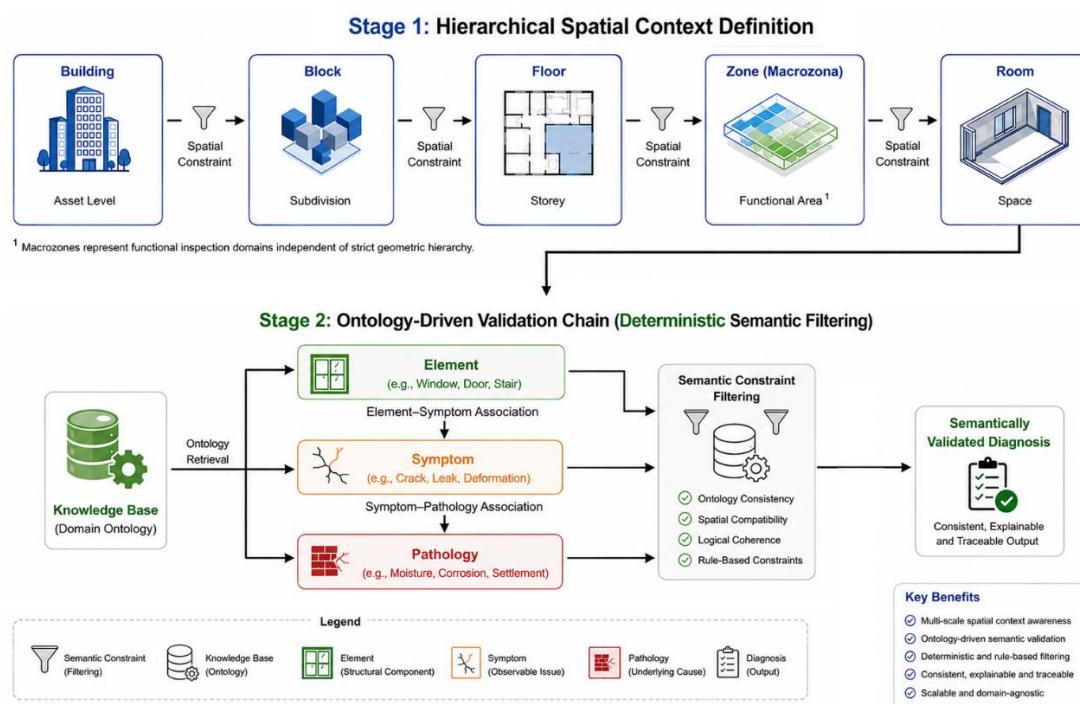


Figure 1: Ontology-driven validation framework integrating hierarchical spatial context definition with a deterministic Element–Symptom–Pathology validation model for building pathology diagnosis.

b) Ontology-Driven Validation

After defining the spatial context, the user proceeds with the diagnostic process:

- The user selects a building Element;
- The system filters and displays only the Symptoms associated with that Element;
- Based on the selected Symptom, the system restricts the available Pathologies to those that are technically correlated.

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This process establishes a deterministic validation chain in which each selection constrains the subsequent one. The integration of spatial context definition with ontology-driven validation ensures that each record is both physically contextualized and semantically consistent. As a result, the system prevents invalid or incompatible combinations, eliminates subjective interpretation at the point of data entry, and guarantees consistency across all inspection records, aligning with integrated facility management approaches that emphasize interoperability and structured data consistency (Eastman et al., 2018).

3.4 Data Integrity Enforcement

The deterministic nature of the validation framework ensures that data integrity is preserved at the point of acquisition. By embedding validation rules directly into the inspection workflow, the system avoids the need for post-processing corrections or data cleansing procedures. All diagnostic records are generated within a constrained semantic environment, where each entry is validated against the ontology structure at the moment of input. This ensures that only technically consistent and semantically valid data is captured.

In addition, the system incorporates a separation between data acquisition and data management layers. The inspection application allows field technicians to submit inspection records but does not provide access to the underlying data repository, nor does it allow modification or editing of submitted records. This architectural separation prevents post-entry manipulation and reinforces the integrity of the dataset, ensuring that recorded information remains an accurate representation of field observations. As a result, the data remains consistent, traceable, and suitable for aggregation and analysis within structured information systems.

3.5 Implementation Context

The proposed validation framework is implemented within the PPMI Digital Twin, a structured information system designed for the management and monitoring of public building assets in the Autonomous Region of the Azores. The system is based on a distributed architecture that separates data acquisition from data storage and management layers.

The inspection process is conducted through a mobile application interface, which operationalizes the ontology-driven validation model and enforces the deterministic workflow defined in the previous sections. Inspection data is captured in the field through a controlled user interface that guides the technician through the spatial context definition and the Element–Symptom–Pathology validation chain. The application ensures that all input is constrained according to predefined semantic rules, preventing inconsistent data entry at the point of acquisition.

Collected data is then stored in a centralized data repository, where it is structured, managed, and made available for analysis within the broader digital environment of the system. This architecture supports data integrity, traceability, and interoperability across multiple system components, consistent with digital twin frameworks that emphasize layered architectures for data integration and information exchange across building systems (de las Morenas et al., 2025).

The integration between the operational inspection interface and the centralized data environment enables a consistent and scalable workflow for data collection and management, reinforcing the reliability of the resulting dataset.

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4. RESULTS

The implementation of the ontology-driven validation framework within the PPMI Digital Twin has resulted in a structured and controlled inspection process, where all diagnostic records are generated under predefined semantic and logical constraints. The results observed are directly related to the enforcement of the deterministic validation model and the integration of spatial context with ontology-based data acquisition.

4.1 Elimination of Invalid Combinations

The deterministic validation mechanism ensures that only technically consistent combinations between Element, Symptom, and Pathology are allowed. As a result, invalid or incompatible associations are systematically prevented at the point of data entry. This eliminates one of the main sources of inconsistency found in traditional inspection processes, where unrestricted input often leads to erroneous or ambiguous diagnostic records.

4.2 Consistency of Diagnostic Records

By constraining user input through a predefined ontology, the system enforces a standardized representation of inspection data. All records follow the same semantic structure, ensuring consistency across different inspections, technicians, and building assets. This uniformity enables reliable comparison between inspection records and supports the aggregation of data at different levels of analysis, consistent with structured facility management approaches supported by integrated information systems (Roper & Payant, 2014).

4.3 Integration of Spatial and Diagnostic Information

The combination of spatial context definition (Building, Block, Floor, Macrozone, Room) with ontology-driven validation ensures that each diagnostic record is both physically contextualized and semantically structured. This integration improves the interpretability of inspection data, allowing each pathology to be precisely located within the building asset while maintaining consistent diagnostic criteria.

4.4 Data Integrity and Traceability

The separation between data acquisition and data management layers prevents modification of inspection records after submission, ensuring that all data remains an accurate representation of field observations. Additionally, the validation rules embedded in the inspection workflow guarantee that all recorded data is compliant with the ontology structure, resulting in datasets that are internally consistent, traceable, and suitable for structured analysis.

4.5 Standardization of Inspection Workflow

The implementation of a guided and constrained inspection process reduces variability in data collection and minimizes dependence on individual interpretation. As a result, the inspection workflow becomes standardized, reproducible, and aligned with defined technical criteria. This standardization contributes to improved data governance and enables the development of scalable information systems for public asset management.

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5. DISCUSSION

The proposed ontology-driven validation framework introduces a structured approach not only to data validation but also to data representation within building inspection processes. A key aspect of the system lies in the definition of the inspection record as an atomic unit of information. Each record corresponds to a single combination of Element–Symptom–Pathology, ensuring semantically precise and technically consistent data entries. This granular structure enables the decomposition of complex inspection scenarios into multiple independent and standardized records.

As a result, a complete inspection of a given space may consist of multiple records, each describing a specific condition associated with a particular building element. This contrasts with traditional inspection methods, where multiple observations are often aggregated into a single, less structured entry. The framework also incorporates practical inspection rules to address situations where no anomalies are detected. In such cases, a representative record is created to reflect the overall condition of space, based on the most representative element. This ensures that all inspected areas are consistently documented, even in the absence of observable pathologies. These principles reinforce both the completeness and consistency of the dataset while maintaining a structured and scalable data model. By defining clear rules for data granularity and representation, the framework supports reliable aggregation and analysis across different inspection contexts. Furthermore, the separation between data acquisition and data management ensures that once records are submitted, they remain immutable. This architectural constraint prevents post-entry modification, reinforcing dataset integrity and enabling full traceability over time.

An additional capability of the framework is the integration of spatial representations of the building asset within the inspection workflow. Following the selection of the spatial context, users are provided with a visual representation of the building layout, enabling intuitive identification of rooms and functional zones. This integration links data acquisition to a graphical representation of the asset, ensuring that inspection records are not only associated with hierarchical spatial attributes but also connected to a visual spatial model. As a result, usability is improved, and the likelihood of location errors during data entry is reduced.

Another important characteristic of the system is the enhanced traceability of inspection records. Each record is uniquely identified, spatially referenced, and associated with a temporal attribute corresponding to the moment of data acquisition. The integration of spatial, semantic, and temporal dimensions establishes a comprehensive data structure that supports both operational control and advanced analytical capabilities. This includes the evaluation of inspection efficiency, comparison of workflows, and future optimization of data acquisition processes. Despite these advantages, the approach introduces certain constraints. The requirement to generate multiple records for a single space may increase data volume, requiring efficient data management strategies. Additionally, the reliance on predefined ontological relationships requires continuous validation and adaptation as new inspection scenarios emerge.

The proposed framework establishes a structured and reproducible paradigm for building pathology diagnosis, significantly improving data integrity and enabling future integration with advanced analytical, machine learning, and intelligent diagnostic systems. The framework is deployed within a BIM 6D environment, supporting integration with lifecycle-oriented asset management processes. By structuring inspection data through a deterministic ontology-based model, the system contributes to the development of consistent and interoperable asset information across different operational stages.

The structured dataset generated by this approach provides a reliable foundation for future integration with machine learning and AI-based diagnostic systems. The consistency, traceability, and semantic clarity of the data enable the development of data-driven analytical models, supporting the transition from reactive inspection practices to predictive and intelligent asset management

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strategies, as highlighted in research on digital twin-enabled predictive maintenance systems (Aivaliotis et al., 2019).

5. Conclusion

This paper introduced a deterministic ontology-driven validation framework for building pathology diagnosis, addressing the fundamental limitations of conventional inspection practices characterized by unstructured data input and subjectivity.

The proposed framework contributes to the field by operationalizing a structured and controlled data acquisition process, ensuring semantic coherence and technical consistency through an ontology-based validation chain. By integrating spatial context with a constrained diagnostic workflow, the approach enables the generation of high-quality, traceable, and interoperable inspection data.

The implementation within a real-world operational environment in the Autonomous Region of the Azores demonstrates the feasibility of the proposed model as a robust data infrastructure for public asset management. The adoption of atomic inspection records, combined with spatial, semantic, and temporal dimensions, establishes a scalable data model that supports consistent information aggregation and lifecycle-oriented asset management processes. In contrast to traditional approaches, the framework enforces data integrity at the point of acquisition, reducing ambiguity and eliminating the need for post-processing validation. This capability aligns with the increasing demand for reliable and structured datasets in Building Information Modelling (BIM) and Digital Twin environments, where data quality is a critical prerequisite for effective integration and decision-making (Boje et al., 2020).

The results indicate that ontology-driven validation can act as an enabling layer for advanced analytical applications. The structured dataset generated through this approach provides a solid foundation for the development of predictive models, automated diagnostics, and intelligent asset management systems.

Overall, this work establishes a reproducible and scalable paradigm for building pathology diagnosis, contributing to the advancement of data-driven asset management and supporting the transition towards integrated Digital Twin ecosystems in the built environment.

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REFERENCES

- Aivaliotis, P., Georgoulas, K., & Chryssolouris, G. (2019). The use of digital twin for predictive maintenance in manufacturing. *International Journal of Computer Integrated Manufacturing*, 32(11), 1067–1081. <https://doi.org/10.1080/0951192X.2019.1686173>
- Boje, C., Guerriero, A., Kubicki, S., & Rezgui, Y. (2020). Towards a semantic Construction Digital Twin: Directions for future research. *Automation in Construction*, 114, 103179. <https://doi.org/10.1016/j.autcon.2020.103179>
- D'Amico, D., Ekoyuncu, J. A., Addepalli, S., Smith, C., Keedwell, E., Sibson, J., & Penver, S. (2020). Conceptual framework of a digital twin to evaluate the degradation status of complex

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engineering systems. *Procedia CIRP*, 86, 61–67.

<https://doi.org/10.1016/j.procir.2020.01.007>

de las Morenas, J., Belmonte, L. M., & Morales, R. (2025). Designing an AI-driven digital twin architecture for building energy prediction. *Journal of Building Engineering*, 99, 111610.

<https://doi.org/10.1016/j.jobbe.2024.111610>

Deng, M., Menassa, C. C., & Kamat, V. R. (2021). From BIM to digital twins: A systematic review of the evolution of intelligent building representations in the AEC-FM industry. *Journal of Information Technology in Construction*, 26, 58–83.

<https://doi.org/10.36680/j.itcon.2021.005>

Roper, K. O., & Payant, R. P. (2014). *The Facility Management Handbook* (4th ed.). AMACOM.

Tao, F., Sui, F., Liu, A., Qi, Q., Zhang, M., Song, B., Guo, Z., Lu, S. C.-Y., & Nee, A. Y. C. (2019). Digital twin-driven product design framework. *International Journal of Production Research*, 57(12), 3935–3953. <https://doi.org/10.1080/00207543.2018.1443229>

Zhao, J., Feng, H., Chen, Q., & Garcia de Soto, B. (2022). Developing a conceptual framework for the application of digital twin technologies to revamp building operation and maintenance processes. *Journal of Building Engineering*, 49, 104028.

<https://doi.org/10.1016/j.jobbe.2022.104028>